강의안 인덱스 (각 섹션 우측 [L##]):
L00 Path Planning 1 | L01 Shortest Path Search | L02 Probabilistic (PRM/RRT) | L03 LPA*/D* Lite | L04 Constraints (nonholonomic, Dubins) | L05 Trajgen & Controllability | L06 Polynomial & Bézier | L07 Collision Check | L08 CHOMP | L09 Drone Corridor | L10 iLQR/LQR | L11 Linear MPC | L12 NMPC/PMP

1. C-space & Completeness [L00,02]

$\mathcal{C}_{\text{obs}} = \{q : A(q) \cap \mathcal{O} \neq \emptyset\}$, $\mathcal{C}_{\text{free}} = \mathcal{C} \setminus \mathcal{C}_{\text{obs}}$. Plan: continuous $\tau : [0,1] \to \mathcal{C}_{\text{free}}$.

- ***Complete***: returns path iff exists.
- ***Resolution complete***: at chosen discretization.
- ***Probabilistic complete***: $P(\text{fail}) \to 0$ as $n \to \infty$.

Minkowski sum: $\mathcal{C}_{\text{obs}} = \mathcal{O} \oplus A$ for point robot.

2. Roadmaps [L00]

Visibility graph: nodes = obstacle vertices + start/goal. Edge if segment is obstacle edge or doesn't cross. $O(n^3)$ naive, $O(n^2)$ optimal.
Voronoi diagram: $\{q : |\text{near}(q)| > 1\}$ — maximally clears obstacles. Straight arc (edge-edge / vertex-vertex), parabolic (edge-vertex). Retraction $\rho : \mathcal{C}_{\text{free}} \to \text{Vor}$.
Cell decomposition: trapezoidal (exact, complete), quadtree/octree (approximate). Build adjacency graph, search.

3. Potential Field [L00]

$U_{\text{att}} = \frac{1}{2}k_a\|x - x_g\|^2$, $F_{\text{att}} = -k_a(x - x_g)$. $F_{\text{rep}} = k_r(\frac{1}{\rho} - \frac{1}{\rho_0})\frac{1}{\rho^2}\nabla\rho$ if $\rho \leq \rho_0$. Issue: local minima. Fix: random walks (RPP), navigation function (no local min, ∞ at obstacles, smooth), wave-front (resolution complete, search from goal).

4. Graph Search [L00,01]

BFS: $O(b^d)$ time/space, complete, optimal (unweighted). DFS: $O(h)$ space, not optimal.
Dijkstra: $f(n) = g(n)$ (no heuristic).
A*: $f(n) = g(n) + h(n)$.

- Optimal iff h ***admissible*** ($h(n) \leq h^*(n)$).
- ***Consistent***: $h(s) \leq c(s, s') + h(s')$, $h(\text{goal}) = 0$.
- Open list (priority queue), Closed list.
- Optimally efficient among same-h algorithms.

Best-first / Greedy: $f = h$. ***Beam search***: keep k best per level. ***RBFS***: DFS-like with f-value backup, $O(bd)$ memory.

5. D*, LPA*, D* Lite [L01,03]

LPA* (incremental A*, fixed start): $rhs(s) = \min_{s' \in \text{Pred}(s)}(g(s') + c(s', s))$, $rhs(s_{\text{start}}) = 0$. ***Locally consistent***: $g = rhs$. Inconsistent in priority queue. Key $[k_1, k_2] = [\min(g, rhs) + h;\ \min(g, rhs)]$. $g > rhs$ (overconsistent): set $g = rhs$. $g < rhs$ (underconsistent): set $g = \infty$.
D* Lite (reverse, moving agent): Search goal→start. $rhs(s) = \min_{s' \in \text{Succ}(s)}(c(s, s') + g(s'))$, $rhs(s_{\text{goal}}) = 0$. Key modifier k_m += $h(s_{\text{last}}, s_{\text{curr}})$ avoids re-sorting queue when robot moves.

6. PRM (multi-query) [L02]

Construction: sample n collision-free q's, connect each to k nearest via local planner (e.g. straight line + incremental/subdivision collision check). **Query**: connect q_i, q_g to roadmap, A*/Dijkstra. Variants: OBPRM (sample near obstacle surfaces), Lazy PRM. Distance: $d(q, q') = w_1\|X - X'\| + w_2 f(R, R')$.

7. RRT & RRT* [L02]

Build: $q_{\text{rand}} \to$ nearest $q_{\text{near}} \to$ step toward $\to q_{\text{new}}$. Add if collision-free. Goal bias 5–10%. **Merge** (bidirectional): grow two trees, alternate. Nodes with largest Voronoi region most likely expanded.

RRT*: Within radius of q_{new}:
1. Choose parent minimizing total cost.
2. ***Rewire***: if going through q_{new} cheaper for q_{near}, change parent.

$\Rightarrow$ probabilistic completeness + ***asymptotic optimality***.

8. Constraints & Nonholonomy [L04]

Pfaffian: $A(q)\dot{q} = 0$, k constraints, $n - k$ DoF velocity. ***Holonomic*** (integrable): can write as $h(q) = 0 \Rightarrow$ lose dimension. ***Nonholonomic***: not integrable; velocity constrained but reachability preserved.
Simple car: $\dot{x} = v\cos\theta$, $\dot{y} = v\sin\theta$, $\dot{\theta} = v\tan\phi/L$, Pfaffian $\dot{x}\sin\theta - \dot{y}\cos\theta = 0$.
Control-affine driftless: $\dot{q} = \sum g_i(q)u_i = G(q)u$.
Lie bracket: $[f, g] = \frac{\partial g}{\partial x}f - \frac{\partial f}{\partial x}g$. Generates new feasible directions over time.
Frobenius: $\mathcal{D} = \text{span}\{g_i\}$ involutive ($[g_i, g_j] \in \mathcal{D}$) $\iff$ holonomic.
Rank test: if $\text{rank}[g_1 \dots g_m\ [g_i, g_j]] > m$ for some $(i, j) \to$ nonholonomic.

9. Controllability [L04,05]

Pfaffian $\to$ ***control system***: $H(q)\dot{q} = 0$ (k rows). Null space basis $\{g_1, \dots, g_{n-k}\} \Rightarrow \dot{q} = G(q)u$, $m = n - k$ inputs. Basis choice 비유일 (다른 u 의미).
Controllable: $\forall q_i, q_f \in \mathcal{C}, \exists u(t)$ steering $q_i \to q_f$. 제약이 도달성에 영향 X.
$\Delta_A = \mathcal{L}(\Delta)$ = involutive closure / ***Lie algebra*** of $\Delta = \text{span}\{g_1, \dots, g_m\}$ = span of all nested Lie brackets:

$$\mathcal{L}(\Delta) = \text{span}\{g_i,\ [g_i, g_j],\ [g_i, [g_j, g_k]], \dots\}$$

$\Delta_i = \Delta_{i-1} + \text{span}\{[g, v] : g \in \Delta_1, v \in \Delta_{i-1}\}$. Accessible dim $\nu = \dim\mathcal{L}(\Delta)$.
왜 *nested?*: 1단 bracket으론 부족할 수 있음.

- Unicycle (n=3,m=2): 1단 $[g_1, g_2]$로 충분.
- Chained-form (n=5,m=2): 1단 rank 3, 2단 4, 3단 5 — 3단까지 필요.
- Snakeboard: drift와 $[f, g_i]$, $[f, [f, g_i]]$ 등 더 깊이.

Termination: 새 bracket이 rank 못 늘리면 stop. $\dim\mathcal{L} \leq n$에서 포화. "No additional vector fields can possibly be independent."
Chow's theorem: $\dim\mathcal{L}(\Delta)(x) = n\,\forall x \iff$ controllable (driftless). Frobenius (1단)로 holonomic 판정 ↔ Chow (전체 nested)로 controllability 판정.
***3-way* 분류** (p = integrable combos, $\nu = n - p$):

case	ν vs n, m	p	성질
완전 nonhol	$\nu = n$	$p = 0$	controllable, STLC
부분 적분	$m < \nu < n$	$0 < p < k$	still nonhol
완전 holonomic	$\nu = m$	$p = k$	integrable

Chow–Rashevskii (driftless): $\dim\mathcal{L}\{g_1, \dots, g_m\} = n \Rightarrow$ STLC. Accessibility $\iff$ controllability.
With drift: accessibility $\not\Rightarrow$ controllability.
Degree of nonholonomy $\kappa \leq n - m + 1$.

10. Dubins / Reeds-Shepp [L04]

Dubins (forward only, min radius ρ): optimal $\in \{LRL, RLR, LSL, LSR, RSL, RSR\}$, $\alpha, \gamma \in [0, 2\pi)$, $\beta \in (\pi, 2\pi)$, $d \geq 0$. CCC if $|q_g - q_i| < 4\rho$.
Reeds-Shepp (forward+reverse): 48 base words including $C|C|C$, $CC|C$, CSC, etc.

11. Chained Form [L05]

2-input, n-state: $\dot{z}_1 = u_1$, $\dot{z}_2 = u_2$, $\dot{z}_3 = z_2u_1, \dots, \dot{z}_n = z_{n-1}u_1$. ***Sufficient conditions***: $\Delta_0 = \text{span}\{g_1, g_2, \text{ad}_{g_1}g_2, \dots, \text{ad}_{g_1}^{n-2}g_2\}$ has dim n; Δ_1, Δ_2 involutive; $\exists h_1$ with $dh_1\Delta_1 = 0$, $dh_1 g_1 = 1$; $\exists h_2$ indep of h_1 with $dh_2\Delta_2 = 0$. Then $z_1 = h_1$, $z_k = L_{g_1}^{n-k}h_2$.
Unicycle: $z_1 = \theta$, $z_2 = x\cos\theta + y\sin\theta$, $z_3 = x\sin\theta - y\cos\theta$. $\omega = v_1$, $v = v_2 + z_3v_1$.
Bicycle (RWD): $z_1 = x$, $z_2 = \sec^3\theta\tan\phi/L$, $z_3 = \tan\theta$, $z_4 = y$. $v = v_1/\cos\theta$, $\omega = -\frac{3}{L}v_1\sec\theta\sin^2\phi + \frac{1}{L}v_2\cos^3\theta\cos^2\phi$. Singular at $\cos\theta = 0$.

12. Differential Flatness [L05,06,09]

$\exists$ flat output y s.t. $x = x(y, \dot{y}, \dots, y^{(r)})$, $u = u(y, \dot{y}, \dots, y^{(r)})$. Driftless: ***flatness*** $\iff$ ***chained form transform***; flat outputs are z_1, z_n.
Unicycle/bicycle: (x, y) flat. $\theta = \arctan(\dot{y}/\dot{x})$, $v = \pm\sqrt{\dot{x}^2 + \dot{y}^2}$, $\omega = (\ddot{y}\dot{x} - \ddot{x}\dot{y})/(\dot{x}^2 + \dot{y}^2)$. ***Quadrotor***: (x, y, z, ψ) flat. $f = m\|a\|$, $\mathbf{b}_3 = a/\|a\|$ with $a = g\mathbf{e}_3 - \ddot{\mathbf{p}}$. Snap $\to$ moments.

13. Polynomial Trajectory [L06]

Cost $J = \int_0^T (P^{(r)}(t))^2 dt$. Euler–Lagrange: $\sum_{k=0}^r (-1)^k \frac{d^k}{dt^k}\frac{\partial L}{\partial x^{(k)}} = 0$. Optimal P: degree $2r - 1$, requires $2r$ BCs.

- $r = 1$ min vel: cubic
- $r = 2$ min accel: quintic
- $r = 3$ min jerk: septic
- $r = 4$ min snap: 9th order

QP form: $\min p^TQp$ s.t. $Ap = b$. With $T^{i+k-2r+1}/(i + k - 2r + 1)$ entries.
Multi-segment: enforce $C^0, \dots, C^{N-1}$ at waypoints, $[A_T^i - A_0^{i+1}][p^i; p^{i+1}] = 0$. Time alloc: $J_{\text{aug}} = J + \sum c_kT_k$.
Quintic Hermite (5th order): $p(u) = 10u^3 - 15u^4 + 6u^5$ for unit BC.
Min jerk (7th): $35u^4 - 84u^5 + 70u^6 - 20u^7$.

14. Bézier / Bernstein [L06,09]

$B_{k,N}(t) = \binom{N}{k}t^k(1 - t)^{N-k}$, $r(t) = \sum b_kB_{k,N}$.

- ***Endpoint interpolation***: $r(0) = b_0$, $r(1) = b_N$.
- ***Convex hull***: $r(t) \in \text{conv}\{b_k\}$ — safety: constrain control points!
- ***Hodograph***: r' is Bézier degree $N - 1$, control points $b_k^{(1)} = N(b_{k+1} - b_k)$. Velocity/accel bounds = bounds on finite diffs of b_k.
- ***Variation diminishing***: curve crosses line control polygon crosses.
- ***Degree elevation***: $\tilde{b}_k = \frac{k}{N+1}b_{k-1} + (1 - \frac{k}{N+1})b_k$.
- ***de Casteljau***: $b_i^k = (1 - t)b_{i-1}^{k-1} + tb_i^{k-1}$. Evaluate + subdivide.

C^0: $P_3 = Q_0$. C^1: $Q_1 = 2P_3 - P_2$. C^2: $Q_2 = 4P_3 - 4P_2 + P_1$.

15. Collision Checking [L07]

Segment intersection (2D): $\text{orient}(A, B, C) = (B_x - A_x)(C_y - A_y) - (B_y - A_y)(C_x - A_x)$. Two segments cross iff orient signs on opposite sides for both pairs.
SAT (convex polygons): for each edge normal u, compute projection intervals; if any separating axis $\to$ no collision.
GJK: $A \cap B \neq \emptyset \iff 0 \in A - B$. Build simplex via support: $\text{support}_{A-B}(d) = \text{support}_A(d) - \text{support}_B(-d)$. Iteratively add support points toward origin; if simplex contains 0 $\to$ collision.
EPA (after GJK collision): expand polytope to find closest boundary point. Penetration depth d + normal n. Recover contact: $v_i = a_i - b_i$, $p_A = \sum\lambda_ia_i$, $p_B = \sum\lambda_ib_i$, $p_A = p_B + dn$.
SDF $\phi(x)$: signed distance (negative inside). Precompute on grid; query forward kinematics $x(q, u)$. Robot $\approx$ sphere: clearance $= \phi(x) - r$.

16. CHOMP [L08]

$\min U(\xi) = f_{\text{prior}}(\xi) + f_{\text{obs}}(\xi)$, $\xi = (q_1, \dots, q_n)$.
Covariant update: $\xi \leftarrow \xi - \frac{1}{\lambda}M^{-1}\nabla U$. M = smoothness metric (PSD).
Smoothness: $f_{\text{prior}} = \frac{1}{2}\sum w_d\|K_d\xi + e_d\|^2$ where K_d = d-th order finite

difference. $= \frac{1}{2}\xi^T A\xi + \xi^T b + c$.

Obstacle gradient (workspace): $\frac{\delta J}{\delta x} = \|x'\|[(I - \hat{x}'\hat{x}'^T)\nabla c - c(x)\kappa]$.
Project to C-space: $\nabla_q f = J_{\text{robot}}^T \nabla_x f$.
Curvature: $\kappa = \frac{1}{\|x'\|^2}(I - \hat{x}'\hat{x}'^T)x''$.
Obstacle cost: $c(x) = \frac{1}{2\epsilon}(d(x) - \epsilon)^2$ for $0 \le d \le \epsilon$, $-d + \frac{\epsilon}{2}$ if $d < 0$, else 0.

HMC inside CHOMP: $p(\xi) \propto \exp(-\alpha U)$, momentum γ, Hamiltonian $H = U + \frac{1}{2}\gamma^T\gamma$. Escape local minima.

17. Safe Flight Corridor [L09]

Given path $\{p_i \to p_{i+1}\}$:

1. ***Ellipsoid*** around segment L_i: start sphere, find nearest obstacle p^*, shrink to touch, repeat.
2. ***Polyhedron***: tangent hyperplanes at obstacle points. $a_j = 2E^{-1}E^{-T}(p_j^c - d)$, $b_j = a_j^T p_j^c$. $C_i = \bigcap H_j$.
3. Optimize Bézier with control points in C_i (convex hull → safety).

FMM (Fast Marching): $|\nabla T| = 1/f(x)$ (Eikonal). $f(d)$ from SDF: low near obstacles. $O(N \log N)$. Path: gradient descent on T.

Cube corridor: axis-aligned inflated cubes — linear constraints $A_i p \le b_i$.

18. LQR & Riccati [L10]

$x_{k+1} = A_k x_k + B_k u_k$, $J = \frac{1}{2}\sum x^T Q x + u^T R u + \frac{1}{2}x_N^T Q_N x_N$. $V_k(x) = \frac{1}{2}x^T P_k x$.

$$K_k = (R + B^T P_{k+1} B)^{-1} B^T P_{k+1} A$$

$$P_k = Q + A^T P_{k+1} A - A^T P_{k+1} B (R + B^T P_{k+1} B)^{-1} B^T P_{k+1} A$$

$P_N = Q_N$. Time-varying linear feedback $u_k^* = -K_k x_k$.

19. iLQR / DDP [L10,12]

Nominal $\{\bar{x}_k, \bar{u}_k\}$. Linearize $\delta x_{k+1} = A_k \delta x_k + B_k \delta u_k$. ***Q-function***: $Q_x = \ell_x + A^T V_x$, $Q_u = \ell_u + B^T V_x$ $Q_{xx} = \ell_{xx} + A^T V_{xx} A$, $Q_{uu} = \ell_{uu} + B^T V_{xx} B$ $Q_{ux} = \ell_{ux} + B^T V_{xx} A$.

Local policy: $\delta u^* = k + K\delta x$, $k = -Q_{uu}^{-1} Q_u$, $K = -Q_{uu}^{-1} Q_{ux}$.

Value update: $V_x = Q_x - Q_{ux}^T Q_{uu}^{-1} Q_u$ $V_{xx} = Q_{xx} - Q_{ux}^T Q_{uu}^{-1} Q_{ux}$.

Forward rollout: $u_k^{\text{new}} = \bar{u}_k + \alpha k_k + K_k(x_k^{\text{new}} - \bar{x}_k)$, line search $\alpha \in (0, 1]$.

Regularization: $Q_{uu}^{\text{reg}} = Q_{uu} + \lambda I$.

DDP: includes f_{xx}, f_{ux}, f_{uu} — extra terms $V_{x,k+1} \cdot F_{**}$ in Q_{**}. iLQR drops them (Gauss-Newton).

20. Linear MPC [L11]

$\min \sum_{i=0}^{N-1}(x^T Q x + u^T R u) + x_N^T P x_N$ s.t. dyn $+\, x \in \mathcal{X}, u \in \mathcal{U}, x_N \in \mathcal{X}_f$. Apply only u_0^*, shift, repeat.

Condensed QP: $X = \Phi x_t + \Gamma U$, $J = U^T H U + 2x^T F^T U +$ const, $H = \Gamma^T \bar{Q}\Gamma + \bar{R}$. Small variable, dense.

Sparse QP: $Z = [x_0; u_0; \ldots; x_N]$, block-banded A_{eq}. Larger but exploits sparsity.

KKT: $\begin{pmatrix} H & A^T \\ A & 0 \end{pmatrix}\begin{pmatrix} Z \\ \lambda \end{pmatrix} = \begin{pmatrix} -q \\ b \end{pmatrix}$.

21. MPC Stability [L11]

Terminal cost V_f approximates V_∞. Choose $K = K_{\text{LQR}}$, $A_K = A + BK$, solve DARE: $P = A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A + Q$.

Terminal set $\mathcal{X}_f$: invariant under K, $K\mathcal{X}_f \subseteq \mathcal{U}$, $\mathcal{X}_f \subseteq \mathcal{X}$, $V_f(A_K x) - V_f(x) \le -\ell(x, Kx)$.

Recursive feasibility: shift optimal sequence + append Kx_N^* remains feasible.

Lyapunov decrease: $V_N(x_{t+1}) - V_N(x_t) \le -\ell(x_t, \kappa_N(x_t))$. Sum to ∞ $\to x_t \to 0$.

Pitfall: terminal cost alone insufficient; need both cost+set.
Soft constraint: $Cx \le d + s$, $s \ge 0$, penalty $\rho_1 \mathbf{1}^T s + \rho_2 s^T s$.

22. NMPC & PMP [L12]

Lagrangian: $\mathcal{L} = \sum \ell + \ell_f + \sum \lambda_k^T (x_{k+1} - F(x_k, u_k))$.

Costate (backward): $\lambda_N = \ell_{f,x}$, $\lambda_k = \ell_x + F_x^T \lambda_{k+1}$.

Optimality: $\ell_u + F_u^T \lambda_{k+1} = 0 \Rightarrow u_k^* = \arg\min_u[\ell + \lambda_{k+1}^T F]$.

SQP: linearize + quadratic cost → QP for $(\delta x, \delta u)$, $\delta x_{k+1} = A_k \delta x_k + B_k \delta u_k$, update $\bar{u}_k \leftarrow \bar{u}_k + \alpha\delta u_k$.

Continuous PMP: $H(x, u, \lambda) = \ell + \lambda^T f$, $\dot{x} = \partial_\lambda H$, $\dot{\lambda} = -\partial_x H$, $u^* = \arg\min_u H$, $\lambda(T) = \partial_x \ell_f$. Two-point BVP (shooting).

Discrete↔PMP: λ_k discrete $= \lambda(t)\Delta t$ continuous, costate recursion matches $\dot{\lambda} = -\partial_x H$ as $\Delta t \to 0$.

23. Stability (discrete) [L11]

$V(x) > 0$, $V(0) = 0$, $\Delta V = V(f(x)) - V(x)$.

- Lyapunov: $\Delta V \le 0$
- Asymptotic: $\Delta V \le -\ell(x)$, $\ell > 0$
- Exponential: $\Delta V \le -\alpha V$

LTI $x_{k+1} = Ax_k$, $\rho(A) < 1$: $V = x^T P x$ where $A^T P A - P = -Q$, $Q > 0$.

24. Method Comparison [전체]

Method	Use case
A*	static, known map
D* Lite	moving agent, changing map
PRM	multi-query, high-D
RRT	single-query, fast
RRT*	asymptotic optimal
CHOMP	local refinement, smooth
SFC+QP	drone, real-time, convex
iLQR	nonlinear, fast, smooth cost
SQP	hard constraints
DDP	2nd-order, more accurate
MPC	online + constraints

Key formulas to remember:

- Pfaffian for car: $\sin\theta\dot{x} - \cos\theta\dot{y} = 0$
- Curvature: $\kappa = (\dot{x}\ddot{y} - \dot{y}\ddot{x})/(\dot{x}^2 + \dot{y}^2)^{3/2}$
- Min-jerk BC: pos+vel+accel+jerk=0 each end
- DARE: $P = A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A + Q$
- iLQR gain: $K = -Q_{uu}^{-1} Q_{ux}$, $k = -Q_{uu}^{-1} Q_u$
- Bézier hodograph: $b_k^{(1)} = N(b_{k+1} - b_k)$
- Hamiltonian: $H = \ell + \lambda^T f$, $\dot{\lambda} = -H_x$